Test Scenarios and Results

Run date 2026-11-02 (run 5, full re-execution from an empty sandbox). Static checks (npm ci, audit, lint, build, Semgrep, secret scan) ran on main HEAD df46295; main HEAD does not build, so behavioural tests (HTTP, browser, axe, ZAP) ran on 1c85d7d, the last buildable commit (the HEAD moved during the run: 14 commits, 23 files). Tools: node 22.23.4, npm 10.9.9, next 14.2.5 (installed), semgrep 1.183.0, OWASP ZAP 2.16.1 (java 21), playwright-core 1.56.1 (pinned), axe-core 4.10.2 (pinned), curl 8.5.0, reportlab 5.0.1.

Tested commit: all results below (HTTP, browser, axe) were executed on commit 1c85d7d, the last buildable commit. Main HEAD df46295 does not build (webpack: getRecordName defined multiple times). A scripted suite run against HEAD with **next dev** returned 10 pass / 86 fail of 96 because the dev server returned HTTP 500 for /, /properties, /kyc/* and /list-your-property; that run is invalid as an app measurement and is kept only as evidence (08b, 08c logs). Code-read deltas on HEAD relevant to failing checks below: /properties and detail pages now have descriptions, canonical (hard-coded https://anarock.com/properties for the list) and Open Graph, /properties adds RealEstateAgent JSON-LD; they are still client-rendered with a Loading shell, the sitemap is unchanged, security headers are unchanged, and /api/ai-search, /api/list-your-property and /api/lead2 still have no rate limit (grep). These HEAD deltas were not executed.

1. Summary

Suite	Executed	Pass	Fail	of which app defect	of which environment / test premise
Scripted HTTP tests (curl, local app, no credentials)	107	64	43	35	7 / 1
Playwright browser tests + axe (desktop 1366 and mobile 375)	112	82	30	10	20 / 0

Run 4 (for comparison): API 49 pass / 35 fail of 84; browser 86 pass / 17 fail of 103. The suites grew (new scenarios: AI search, list-your-property, /aboutus, SoW pages) so totals are not directly comparable. Environmental failures are caused by the sandbox (no credentials by design, proxy-blocked outbound HTTPS) and are **not** reported as app failures.

2. Scripted HTTP results

Result	Group	Scenario	Class / reason
PASS	PG	GET /	
PASS	PG	GET /aboutus	
PASS	PG	GET /properties	
PASS	PG	GET /compare	
PASS	PG	GET /shortlist	
PASS	PG	GET /services	
PASS	PG	GET /list-your-property	
PASS	PG	GET /kyc/city/delhi	
PASS	PG	GET /kyc/city/mumbai	
PASS	PG	GET /kyc/city/bengaluru	
PASS	PG	GET /kyc/city/pune	
PASS	PG	GET /kyc/city/chennai	
PASS	PG	GET /kyc/city/hyderabad	
PASS	PG	GET /kyc/city/kolkata	
PASS	PG	GET /kyc/city/noida	
PASS	PG	GET /kyc/city/gurugram	
PASS	PG	GET /unknown-route	

Result	Group	Scenario	Class / reason
FAIL	PG	GET /properties/abc/xyz (non-existent property) should be 404	App defect
PASS	SEO	sitemap URL resolves: /	
PASS	SEO	sitemap URL resolves: /properties	
PASS	SEO	sitemap URL resolves: /kyc/city/bengaluru	
PASS	SEO	sitemap URL resolves: /kyc/city/chennai	
PASS	SEO	sitemap URL resolves: /kyc/city/delhi	
PASS	SEO	sitemap URL resolves: /kyc/city/gurugram	
PASS	SEO	sitemap URL resolves: /kyc/city/hyderabad	
PASS	SEO	sitemap URL resolves: /kyc/city/kolkata	
PASS	SEO	sitemap URL resolves: /kyc/city/mumbai	
PASS	SEO	sitemap URL resolves: /kyc/city/noida	
PASS	SEO	sitemap URL resolves: /kyc/city/pune	
FAIL	SEO	sitemap contains property/detail URLs	App defect sitemap.js lists only static pages and city pages (code read); it never fetches property URLs, with or without data.
FAIL	SEO	sitemap domain equals request host (NEXT_PUBLIC_BASE_URL set)	Test premise NEXT_PUBLIC_BASE_URL is inlined at build time and the build ran before the env var was set; test premise invalid, inconclusive.
PASS	SEO	robots.txt references sitemap	
FAIL	SEO	robots.txt disallow paths exist as routes (/wishlist)	App defect
PASS	SEO	home has <title>	
PASS	SEO	home has name="description"	
PASS	SEO	home has rel="canonical"	
PASS	SEO	home has property="og:title"	
PASS	SEO	home has name="twitter:card"	
PASS	SEO	home has application/ld+json	
FAIL	SEO	/properties HTML is server-rendered with listings (not 'Loading properties')	App defect
PASS	SEO	property detail has its own meta description (differs from home)	
FAIL	SEO	property detail JSON-LD is a listing type (RealEstateListing/Product/Offer)	App defect
FAIL	SEO	canonical of /properties points to itself (not homepage)	App defect
FAIL	SEO	canonical of /properties/1/test-property points to itself (not homepage)	App defect
FAIL	SEO	canonical of /services points to itself (not homepage)	App defect
FAIL	SEO	canonical of /kyc/city/delhi points to itself (not homepage)	App defect
FAIL	SEO	canonical of /list-your-property points to itself (not homepage)	App defect
FAIL	HDR	header present: content-security-policy	App defect
FAIL	HDR	header present: strict-transport-security	App defect
FAIL	HDR	header present: x-frame-options	App defect
FAIL	HDR	header present: x-content-type-options	App defect
FAIL	HDR	header present: referrer-policy	App defect
FAIL	HDR	header present: permissions-policy	App defect

Result	Group	Scenario	Class / reason
FAIL	HDR	X-Powered-By not exposed	App defect
PASS	XSS	slug with HTML is escaped in <title>	
PASS	XSS	/api/wishlist reflects sessionId only as JSON	
PASS	LEAD	missing name	
PASS	LEAD	invalid phone	
PASS	LEAD	invalid email	
PASS	LEAD	wrong content-type	
FAIL	LEAD	malformed JSON should be 400	App defect
PASS	LEAD	first name > 80 chars rejected	
PASS	LEAD	60KB body with Content-Length rejected	
FAIL	LEAD	60KB chunked body (no Content-Length) should still be rejected	App defect
PASS	LEAD	honeypot returns 200 (silent) without CRM call	
PASS	LEAD	DELETE not allowed	
FAIL	LEAD2	invalid phone/email rejected with 400 (lead2)	Environment Without credentials /api/lead2 fails at the Zoho token step before returning validation results (generic 500/LEAD_SUBMISSION_FAILED); code read shows it has no content-type, size or phone/email validation (finding S-02).
FAIL	LEAD2	text/plain rejected with 415 (lead2)	Environment Without credentials /api/lead2 fails at the Zoho token step before returning validation results (generic 500/LEAD_SUBMISSION_FAILED); code read shows it has no content-type, size or phone/email validation (finding S-02).
FAIL	LEAD2	60KB body (valid name/phone) rejected with 413 (lead2)	Environment Without credentials /api/lead2 fails at the Zoho token step before returning validation results (generic 500/LEAD_SUBMISSION_FAILED); code read shows it has no content-type, size or phone/email validation (finding S-02).
PASS	LEAD	60KB body (valid name/phone) rejected with 413 (/api/leads, control)	
FAIL	LEAD2	cross-origin text/plain POST (no CORS preflight) rejected	Environment Without credentials /api/lead2 fails at the Zoho token step before returning validation results (generic 500/LEAD_SUBMISSION_FAILED); code read shows it has no content-type, size or phone/email validation (finding S-02).
PASS	LEAD3	invalid phone rejected (lead3)	
PASS	RATE	/api/leads same client: 11th request is 429	
FAIL	RATE	/api/leads rotating X-Forwarded-For still limited	App defect
PASS	RATE	/api/lead3 same client: 11th request is 429	
FAIL	RATE	/api/lead3 rotating X-Forwarded-For still limited	App defect
FAIL	RATE	/api/lead2 same client: limited	App defect
FAIL	PG	SoW 5.1 static page exists: GET /privacy	App defect
FAIL	PG	SoW 5.1 static page exists: GET /terms	App defect
FAIL	PG	SoW 5.1 static page exists: GET /contact	App defect
FAIL	PG	stray /HomePage route (pages/ router) should not be publicly routable	App defect
PASS	AI	ai-search empty prompt	
FAIL	AI	ai-search malformed JSON	App defect
FAIL	AI	ai-search malformed JSON error body must not leak parser internals	App defect
FAIL	AI	ai-search oversized prompt (20KB) rejected 400/413	App defect

Result	Group	Scenario	Class / reason
FAIL	RATE	/api/ai-search (paid Gemini call) is rate limited: 429 within 15 requests from one client	App defect
PASS	LYP	list-your-property POST without multipart	
FAIL	RATE	/api/list-your-property is rate limited: 429 within 15 requests	App defect
PASS	API	properties?limit=100000 is capped (response < 5MB)	
FAIL	SEC	Server header/X-Powered-By version not exposed	App defect
PASS	SEC	TRACE method not allowed on /	
PASS	SEC	unknown /api route returns JSON/404 not stack trace	
FAIL	API	exchange-rates 200	Environment Outbound HTTPS from the app is blocked by the sandbox proxy (SELF_SIGNED_CERT_IN_CHAIN in server log); not an app failure.
FAIL	API	exchange-rates has INR,AED,USD,EUR,SGD	Environment Outbound HTTPS from the app is blocked by the sandbox proxy (SELF_SIGNED_CERT_IN_CHAIN in server log); not an app failure.
PASS	API	kyc without city	
PASS	API	wishlist without sessionId	
PASS	API	location lat out of range	
FAIL	API	location malformed JSON	App defect
FAIL	API	location valid coordinates (New Delhi)	Environment Outbound HTTPS from the app is blocked by the sandbox proxy (SELF_SIGNED_CERT_IN_CHAIN in server log); not an app failure.
FAIL	API	micromarkets upstream failure is not reported as success	App defect
PASS	API	/api/properties error body must not disclose env var names	
FAIL	API	/api/cities error body must not disclose env var names	App defect
PASS	CORS	cross-origin GET to /api/properties does not echo Access-Control-Allow-Origin: *	
PASS	CORS	OPTIONS /api/leads does not allow arbitrary origin	
PASS	ERR	/api/properties/{id} error body must not disclose env var names	
PASS	ERR	/api/kyc?city=delhi error body must not disclose env var names	
PASS	ERR	path traversal in slug does not 500	

3. Browser results (Playwright 1.56.1 + axe-core 4.10.2; desktop then mobile)

Result	Group	Scenario	Class / reason
PASS	PAGE	/ loads (HTTP 200, 8222ms)	
PASS	LAYOUT	/ no horizontal scroll	
FAIL	JS	/ no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/ does not show internal env-var errors to the user	
FAIL	A11Y	/ axe: no critical/serious WCAG2.1 A/AA violations	App defect [aria-required-parent(critical,2), color-contrast(serious,3) | all: aria-required-parent:2,color-contrast:3]
FAIL	PAGE	/aboutus loads (HTTP ERR, 45003ms)	Environment networkidle never reached in 45 s although curl returns HTTP 200; most likely blocked external assets in the sandbox. Unverified.
PASS	LAYOUT	/aboutus no horizontal scroll	
FAIL	JS	/aboutus no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/aboutus does not show internal env-var errors to the user	
PASS	A11Y	/aboutus axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/properties loads (HTTP 200, 676ms)	
PASS	LAYOUT	/properties no horizontal scroll	
FAIL	JS	/properties no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/properties does not show internal env-var errors to the user	
FAIL	A11Y	/properties axe: no critical/serious WCAG2.1 A/AA violations	App defect [select-name(critical,2) | all: select-name:2]
PASS	PAGE	/shortlist loads (HTTP 200, 594ms)	
PASS	LAYOUT	/shortlist no horizontal scroll	
FAIL	JS	/shortlist no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/shortlist does not show internal env-var errors to the user	
PASS	A11Y	/shortlist axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/compare loads (HTTP 200, 613ms)	
PASS	LAYOUT	/compare no horizontal scroll	
FAIL	JS	/compare no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/compare does not show internal env-var errors to the user	
PASS	A11Y	/compare axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/list-your-property loads (HTTP 200, 657ms)	
PASS	LAYOUT	/list-your-property no horizontal scroll	
FAIL	JS	/list-your-property no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/list-your-property does not show internal env-var errors to the user	
FAIL	A11Y	/list-your-property axe: no critical/serious WCAG2.1 A/AA violations	App defect [select-name(critical,2) | all: select-name:2]
PASS	PAGE	/services loads (HTTP 200, 607ms)	

Result	Group	Scenario	Class / reason
PASS	LAYOUT	/services no horizontal scroll	
FAIL	JS	/services no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/services does not show internal env-var errors to the user	
PASS	A11Y	/services axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/kyc/city/delhi loads (HTTP 200, 615ms)	
PASS	LAYOUT	/kyc/city/delhi no horizontal scroll	
FAIL	JS	/kyc/city/delhi no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/kyc/city/delhi does not show internal env-var errors to the user	
FAIL	A11Y	/kyc/city/delhi axe: no critical/serious WCAG2.1 A/AA violations	App defect [color-contrast(serious,2) | all: color-contrast:2]
PASS	PAGE	/properties/1/test-property loads (HTTP 200, 609ms)	
PASS	LAYOUT	/properties/1/test-property no horizontal scroll	
FAIL	JS	/properties/1/test-property no unexpected console/page errors	Environment No Catalyst credentials by design (nothing may reach production), so listing data is unavailable. [Error: Properties data is temporarily unavailable. Retry after 55 seconds.]
PASS	UI	/properties/1/test-property does not show internal env-var errors to the user	
PASS	A11Y	/properties/1/test-property axe: no critical/serious WCAG2.1 A/AA violations	
PASS	LINK	GET /	
PASS	LINK	GET /properties	
PASS	LINK	GET /aboutus	
PASS	LINK	GET /#enquiry	
PASS	LINK	GET /list-your-property	
PASS	LINK	GET /shortlist	
PASS	LINK	GET /kyc/city/bengaluru	
PASS	LINK	GET /kyc/city/chennai	
PASS	LINK	GET /kyc/city/delhi	
PASS	LINK	GET /kyc/city/gurugram	
PASS	LINK	GET /kyc/city/hyderabad	
PASS	LINK	GET /kyc/city/kolkata	
PASS	LINK	GET /kyc/city/mumbai	
PASS	LINK	GET /kyc/city/noida	
PASS	LINK	GET /kyc/city/pune	
FAIL	LINK	GET /privacy-policy	App defect [404]
FAIL	LINK	GET /rera	App defect [404]
FAIL	LINK	GET /terms-of-use	App defect [404]
PASS	SEC	target=_blank links have rel=noopener	
PASS	COOKIE	consent banner visible on first visit	
PASS	COOKIE	no cookies set before consent	

Result	Group	Scenario	Class / reason
PASS	FORM	a lead CTA button exists on home	
PASS	PAGE	/ loads (HTTP 200, 2168ms)	
PASS	LAYOUT	/ no horizontal scroll	
FAIL	JS	/ no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/ does not show internal env-var errors to the user	
FAIL	A11Y	/ axe: no critical/serious WCAG2.1 A/AA violations	App defect [aria-required-parent(critical,2), color-contrast(serious,3) | all: aria-required-parent:2,color-contrast:3]
FAIL	PAGE	/aboutus loads (HTTP ERR, 45002ms)	Environment networkidle never reached in 45 s although curl returns HTTP 200; most likely blocked external assets in the sandbox. Unverified.
PASS	LAYOUT	/aboutus no horizontal scroll	
FAIL	JS	/aboutus no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/aboutus does not show internal env-var errors to the user	
PASS	A11Y	/aboutus axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/properties loads (HTTP 200, 647ms)	
PASS	LAYOUT	/properties no horizontal scroll	
FAIL	JS	/properties no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/properties does not show internal env-var errors to the user	
PASS	A11Y	/properties axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/shortlist loads (HTTP 200, 586ms)	
PASS	LAYOUT	/shortlist no horizontal scroll	
FAIL	JS	/shortlist no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/shortlist does not show internal env-var errors to the user	
PASS	A11Y	/shortlist axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/compare loads (HTTP 200, 575ms)	
PASS	LAYOUT	/compare no horizontal scroll	
FAIL	JS	/compare no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/compare does not show internal env-var errors to the user	
PASS	A11Y	/compare axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/list-your-property loads (HTTP 200, 615ms)	
PASS	LAYOUT	/list-your-property no horizontal scroll	
FAIL	JS	/list-your-property no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/list-your-property does not show internal env-var errors to the user	
FAIL	A11Y	/list-your-property axe: no critical/serious WCAG2.1 A/AA violations	App defect [select-name(critical,2) | all: select-name:2]
PASS	PAGE	/services loads (HTTP 200, 610ms)	
PASS	LAYOUT	/services no horizontal scroll	

Result	Group	Scenario	Class / reason
FAIL	JS	/services no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/services does not show internal env-var errors to the user	
PASS	A11Y	/services axe: no critical/serious WCAG2.1 A/AA violations	
PASS	PAGE	/kyc/city/delhi loads (HTTP 200, 598ms)	
PASS	LAYOUT	/kyc/city/delhi no horizontal scroll	
FAIL	JS	/kyc/city/delhi no unexpected console/page errors	Environment Page calls /api/exchange-rates, which cannot reach its external provider from this sandbox. [Exchange rate loading failed: Error: Unable to fetch exchange rates.]
PASS	UI	/kyc/city/delhi does not show internal env-var errors to the user	
FAIL	A11Y	/kyc/city/delhi axe: no critical/serious WCAG2.1 A/AA violations	App defect [color-contrast(serious,2) | all: color-contrast:2]
PASS	PAGE	/properties/1/test-property loads (HTTP 200, 584ms)	
PASS	LAYOUT	/properties/1/test-property no horizontal scroll	
FAIL	JS	/properties/1/test-property no unexpected console/page errors	Environment No Catalyst credentials by design (nothing may reach production), so listing data is unavailable. [Error: Properties data is temporarily unavailable. Retry after 60 seconds.]
PASS	UI	/properties/1/test-property does not show internal env-var errors to the user	
PASS	A11Y	/properties/1/test-property axe: no critical/serious WCAG2.1 A/AA violations	

4. Designed scenarios that could not be executed

Scenario	Status	Reason
Lead submitted from home/property forms creates a CRM Lead end to end (incl. duplicate handling, owner assignment)	Blocked	Writing to production CRM is prohibited; no CRM sandbox exists. Validation-only paths were tested.
Listing, filter, detail, compare and wishlist with real property data and Stratus images	Blocked	No Catalyst credentials; app run without them by design.
AI-assisted search returns ranked results from Gemini 2.5 Flash	Blocked	No GEMINI_API_KEY; outbound HTTPS blocked in sandbox. Only input handling tested.
Currency (INR, AED, USD, EUR, SGD) and unit switching with live rates	Blocked	/api/exchange-rates cannot reach its provider from the sandbox (502); code supports the five currencies and sq ft / sq m.
Reverse geocoding of 'use my location'	Blocked	Nominatim unreachable from the sandbox.
Property approval in CRM -> images to Stratus -> Gemini attributes -> Data Store -> revalidation	Blocked	No such Function is deployed or in the repo; cannot be executed.
SalesIQ chat, chatbot handover, PageSense heatmaps, GA4 events, view-count scoring	Blocked	Features do not exist in the code (Scope report).
On-demand revalidation and scheduled sitemap regeneration	Blocked	No revalidate code or cron job exists.
Production site over HTTPS on the custom domain (headers, TLS, CDN caching, Core Web Vitals)	Blocked	Production Slate deployment failed; no custom domain linked.
Active DAST (ZAP active scan, authenticated tests)	Blocked	Deliberately not run; no authenticated surface exists.